AN OBJECT OF THE DERIVED CATEGORY IS INVERTIBLE EXACTLY WHEN IT IS LOCALLY A SHIFT OF THE RING, AND THEN ITS COHOMOLOGY IS FINITE PROJECTIVE IN A SINGLE DEGREE PER FACTOR

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

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Editorial note. The author writes 'we omit the details' at proof line 38682 for the local checking of the composition map; the phrase is kept verbatim. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Invertible objects in a symmetric monoidal category and their left duals. condition (1) at statement lines 38661-38662 and used at proof lines 38683 and 38686-38688. Is the notion at issue and the source of the left dual that gives perfection.
- The residue-field fibre M tensor^L_R kappa and its cohomology dimensions. used at proof lines 38689-38693. Forces the single-degree concentration that characterises condition (2).
- Tor amplitude of a perfect complex and the product decomposition of R over disjoint open subsets of the spectrum. used at proof lines 38701-38705. Produces the global product decomposition R = prod R_n and hence the remaining assertions.

Source reference. Original lemma 00EM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 540–540. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0FFN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19293–19293. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

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Source reference. Original lemma 38581: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38581–38581. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 19594: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19594–19594. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let M be an object of $D(R)$. The following are equivalent*

(1) *M is invertible in $D(R)$, see Categories, Definition 0FFN, and*
(2) *for every prime ideal $\mathfrak{p} \subset R$ there exists an $f \in R$, $f \notin \mathfrak{p}$ such that $M_f \cong R_f[-n]$ for some $n \in \mathbf{Z}$.*

Moreover, in this case

(a) *M is a perfect object of $D(R)$,*
(b) *$M = \bigoplus H^n(M)[-n]$ in $D(R)$,*
(c) *each $H^n(M)$ is a finite projective R-module,*
(d) *we can write $R = \prod_{a \leq n \leq b} R_n$ such that $H^n(M)$ corresponds to an invertible R_n-module.*

Proof. Assume (2). Consider the object $R\mathrm{Hom}_R(M, R)$ and the composition map

$$R\mathrm{Hom}(M, R) \otimes^{\mathbf{L}}_R M \to R$$

Checking locally we see that this is an isomorphism; we omit the details. Because $D(R)$ is symmetric monoidal we see that M is invertible.

Assume (1). Observe that an invertible object of a monoidal category has a left dual, namely, its inverse. Thus M is perfect by Lemma 38581. Consider a prime ideal $\mathfrak{p} \subset R$ with residue field κ. Then we see that $M \otimes^{\mathbf{L}}_R \kappa$ is an invertible object of $D(\kappa)$. Clearly this implies that $\dim H^i(M \otimes^{\mathbf{L}}_R \kappa)$ is nonzero exactly for one i and equal to 1 in that case. By Lemma 19594 this gives (2).

In the proof above we have seen that (a) holds. Let $U_n \subset \mathrm{Spec}(R)$ be the union of the opens of the form $D(f)$ such that $M_f \cong R_f[-n]$. Clearly, $U_n \cap U_{n'} = \emptyset$ if $n \neq n'$. If M has tor amplitude in $[a, b]$, then $U_n = \emptyset$ if $n \notin [a, b]$. Hence we see that we have a product decomposition $R = \prod_{a \leq n \leq b} R_n$ as in (d) such that U_n corresponds to $\mathrm{Spec}(R_n)$, see Algebra, Lemma 00EM. Since $D(R) = \prod_{a \leq n \leq b} D(R_n)$ and similarly for the category of modules parts (b), (c), and (d) follow immediately. □